

t = 0,1,2....9

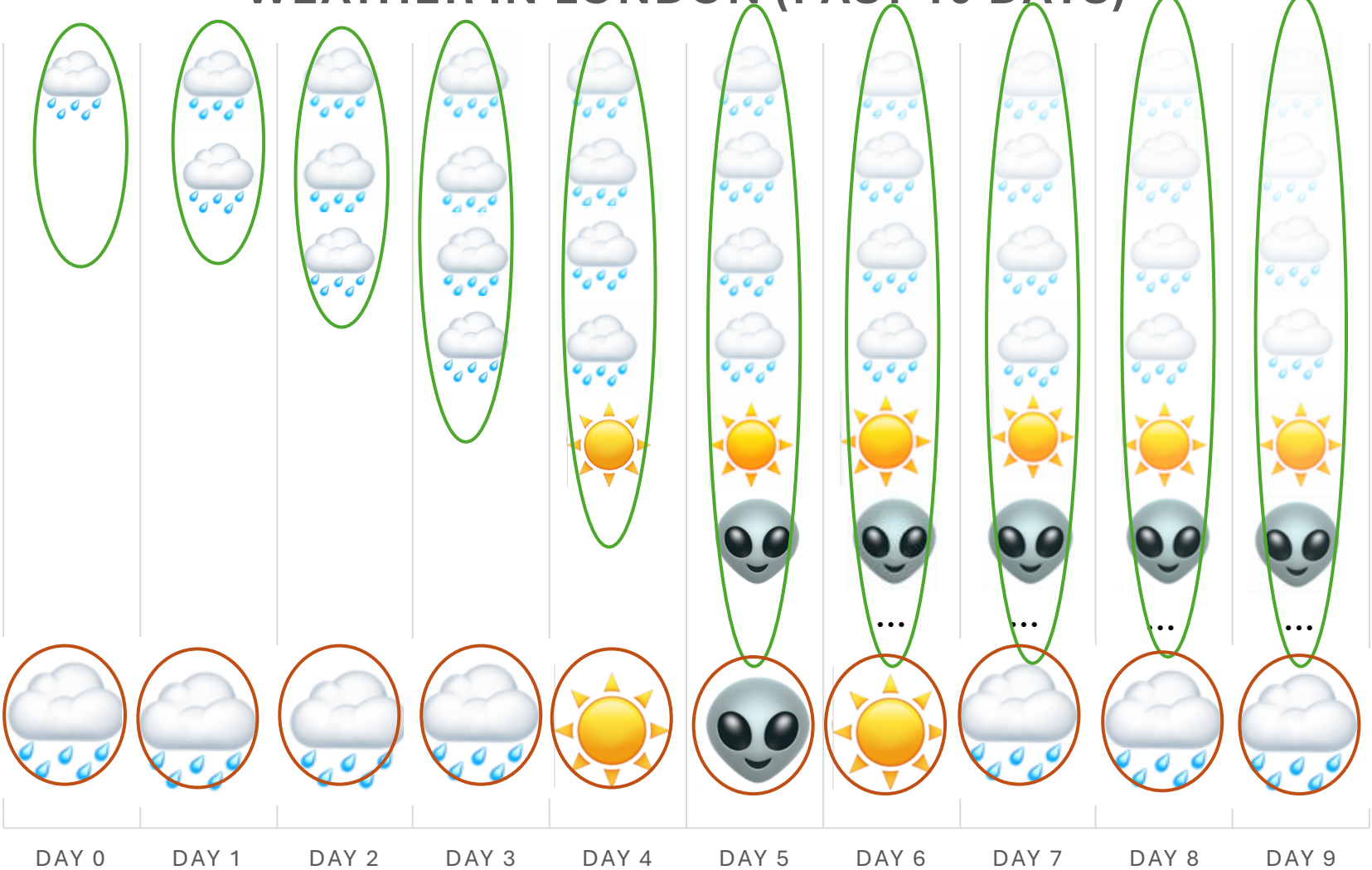
h_t

Memory
till day t

x_t

Weather
at day t

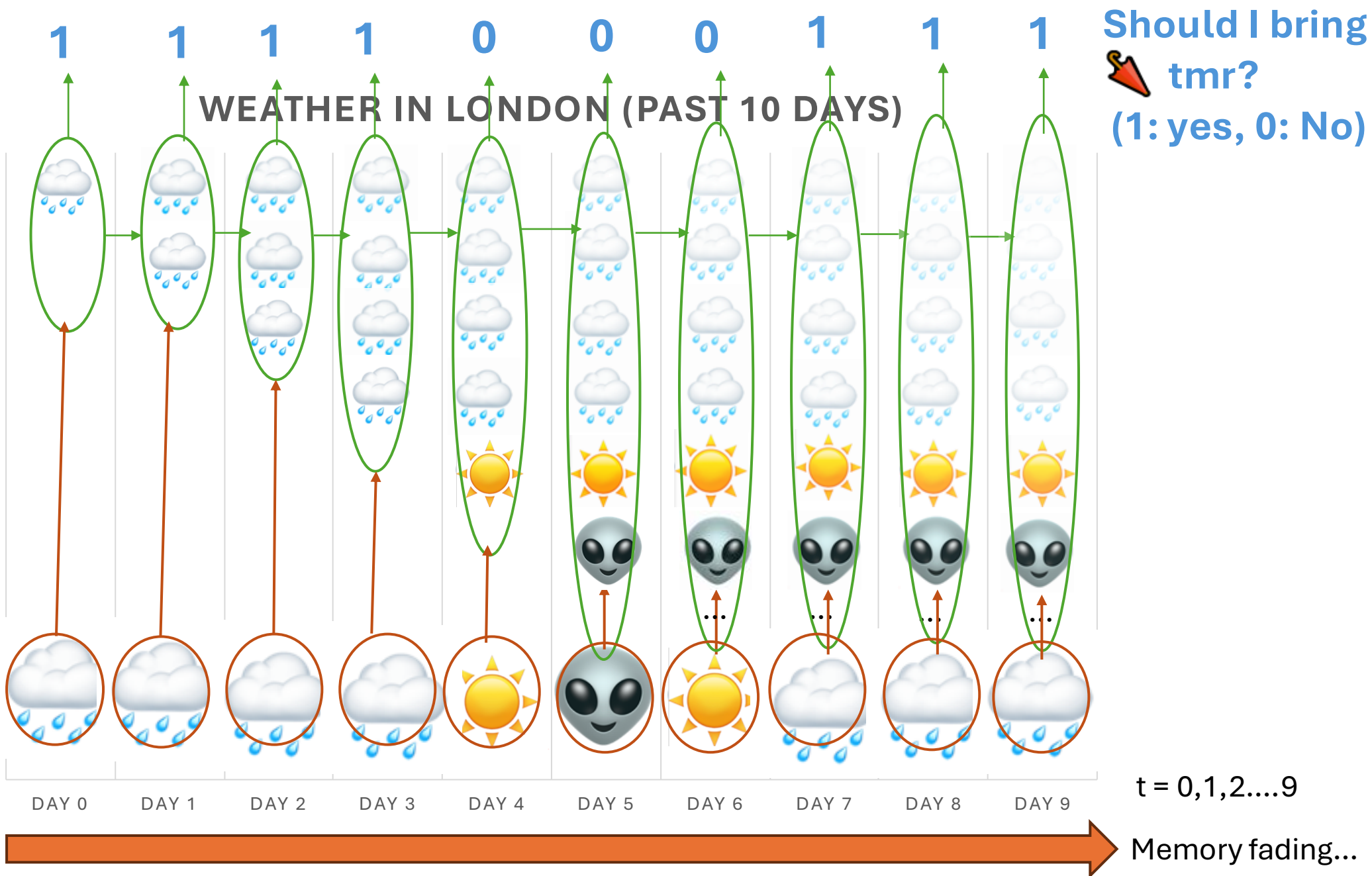
WEATHER IN LONDON (PAST 10 DAYS)



y_t
Output
at day t

h_t
Memory
till day t

x_t
Weather
at day t



It is naturally to think that:

$$h_t = f(h_{t-1}, x_t), \quad y_t = g(h_t)$$



$$a, b, c \in \mathbb{R}, \quad h_t, x_t, y_t \in \mathbb{R} \text{ for all } t.$$

$$h_t = a h_{t-1} + b x_t, \quad y_t = c h_t$$

Remark: Here a, b, c could be interpreted as “weight”.

Of course, 1-D is not enough, let's expand the system to higher dimensions.

Here, our capacity of memories (d_{states}) and the feature dim for input (d_{model}).

$$h_t = a h_{t-1} + b x_t, \quad y_t = c h_t$$



Expand to $h_t \in \mathbb{R}^{d_{\text{state}}}$, $x_t, y_t \in \mathbb{R}^{d_{\text{model}}}$ for all t .

$$h_t = A h_{t-1} + B x_t, \quad y_t = C h_t$$



LTI (A, B, C) $\rightarrow$ time-dependent (A_t, B_t, C_t).

$$h_t = A_t h_{t-1} + B_t x_t, \quad y_t = C_t h_t$$

Remark: In neuroscience terms, the brain gates memory. salient or novel events (👁️) trigger stronger encoding or even a reset of what's stored, rather than a fixed-rate write/forget.

That exactly what Mamba trying to do. That's why we called it a Selective SSM.

$$\dot{h}(t) = Ah(t) + B_t x(t), \quad y(t) = C_t h(t).$$



ZOH discretisation with step size Δ_t .

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t$$

$$\bar{A}_t = \exp(\Delta_t A), \quad \bar{B}_t = (\Delta_t A)^{-1} (\exp(\Delta_t A) - I) \Delta_t B_t.$$

Remark: when dt increases, mentally speaking dtA decreases then $\exp(dtA)$ decreases, means we give our past little weight (tend to forget more); also $(I - \exp(dtA))$ increases, means we give more weight to input. Therefore, dt somehow controls the trade-off between how fast we forget and how much new things to write in.

$$\Delta_t = f(x_t), \quad B_t = g(x_t), \quad C_t = h(x_t)$$

$$\Rightarrow \bar{A}_t = \exp(\Delta_t A) \text{ is time-/input-dependent}$$

In our London weather story:

Small dt case:

x_t



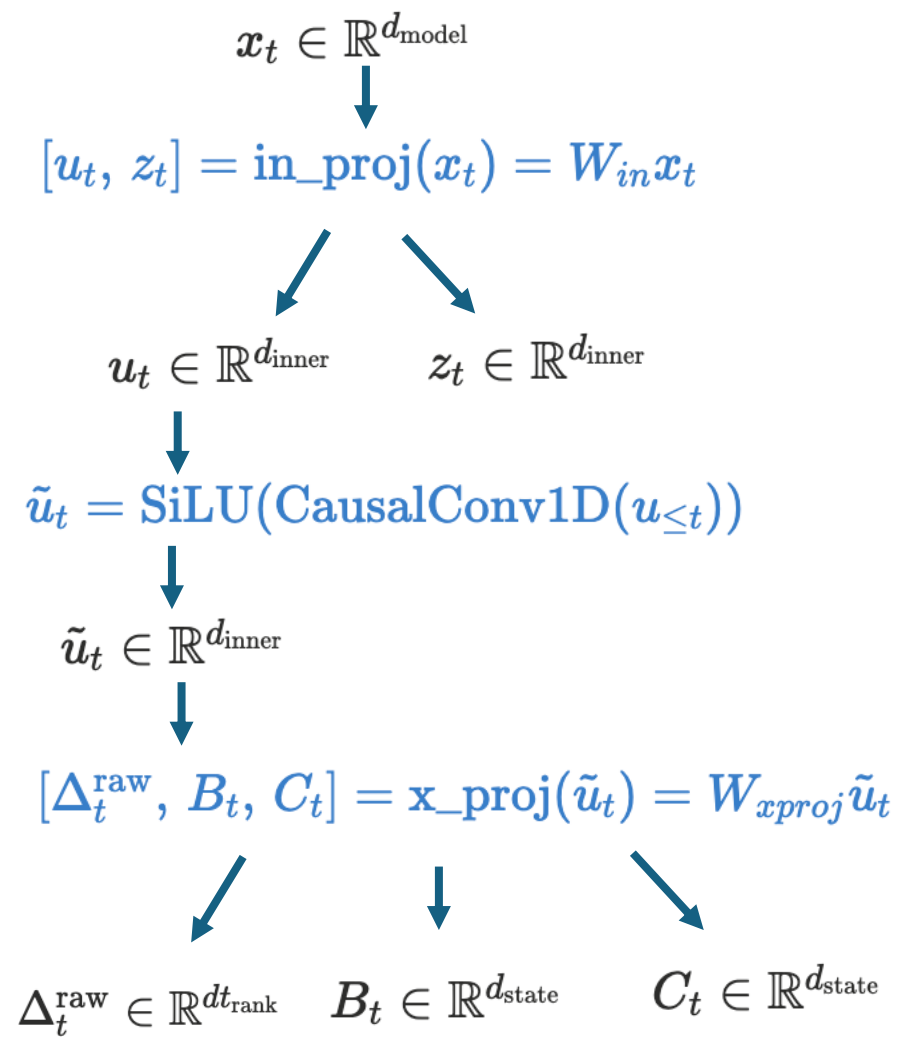
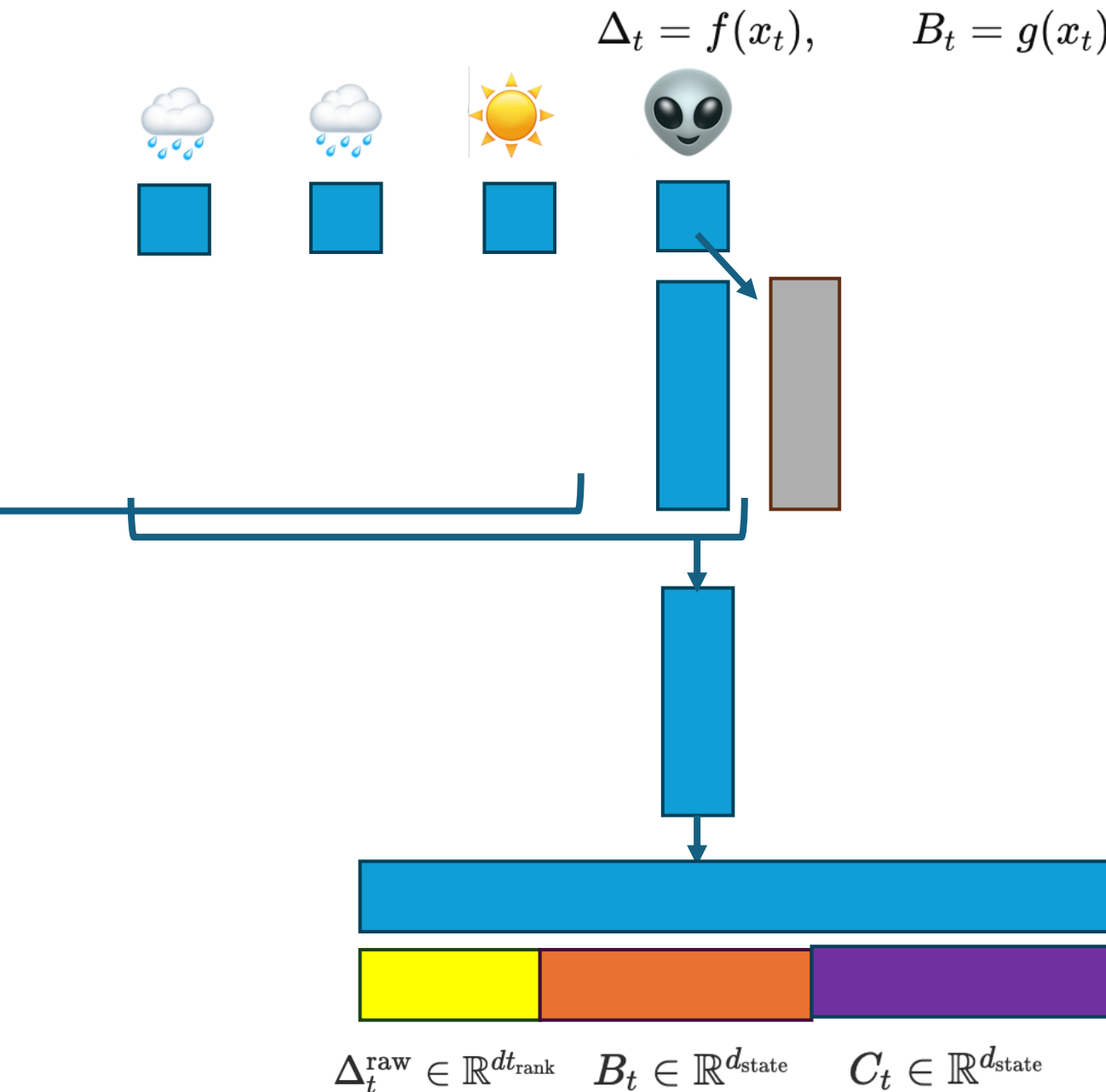
$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$$

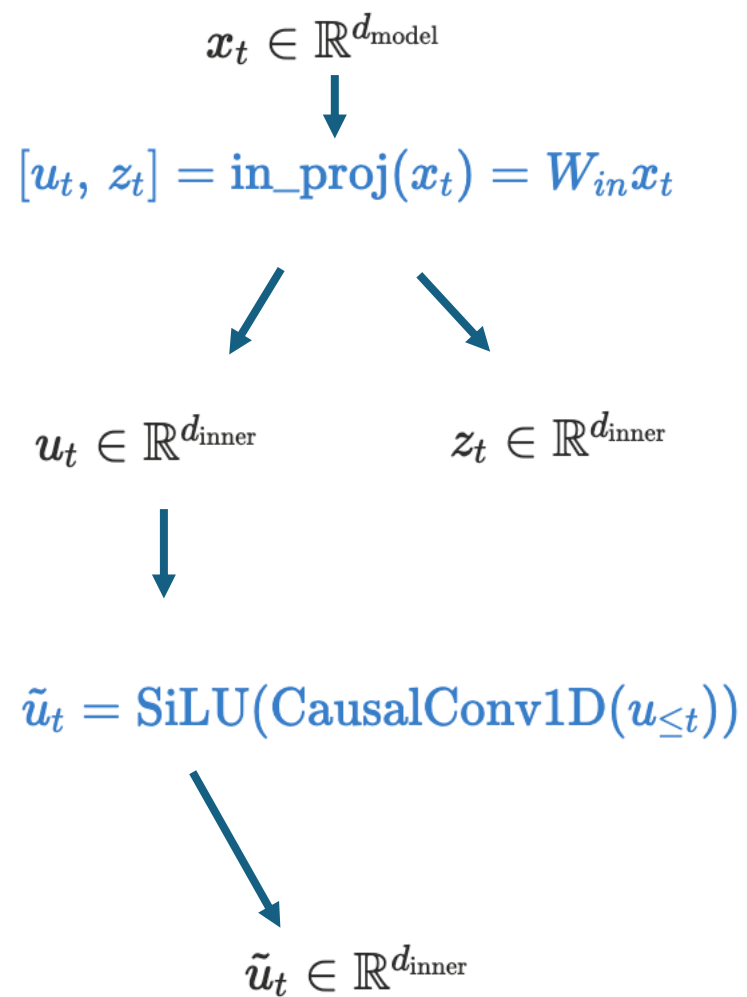
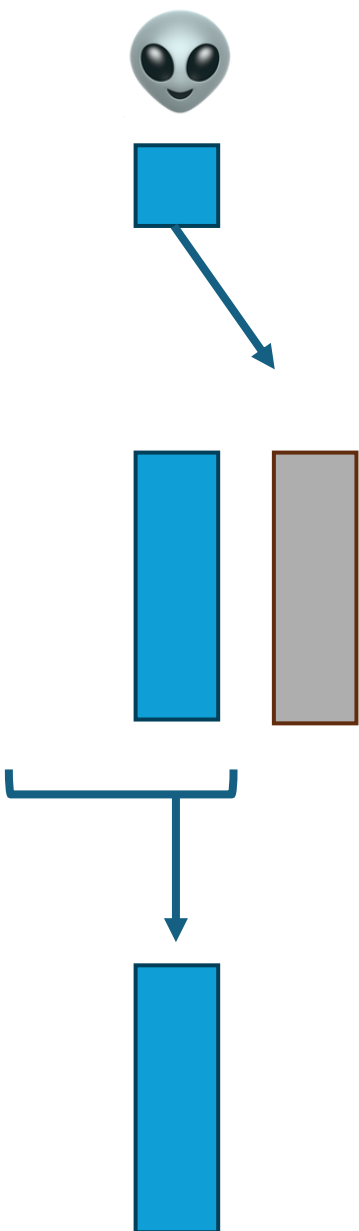
Large dt case:



$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$$

Now we are interested how we have Δ_t , B_t , C_t from x_t . We want to write out f , g , h explicitly.

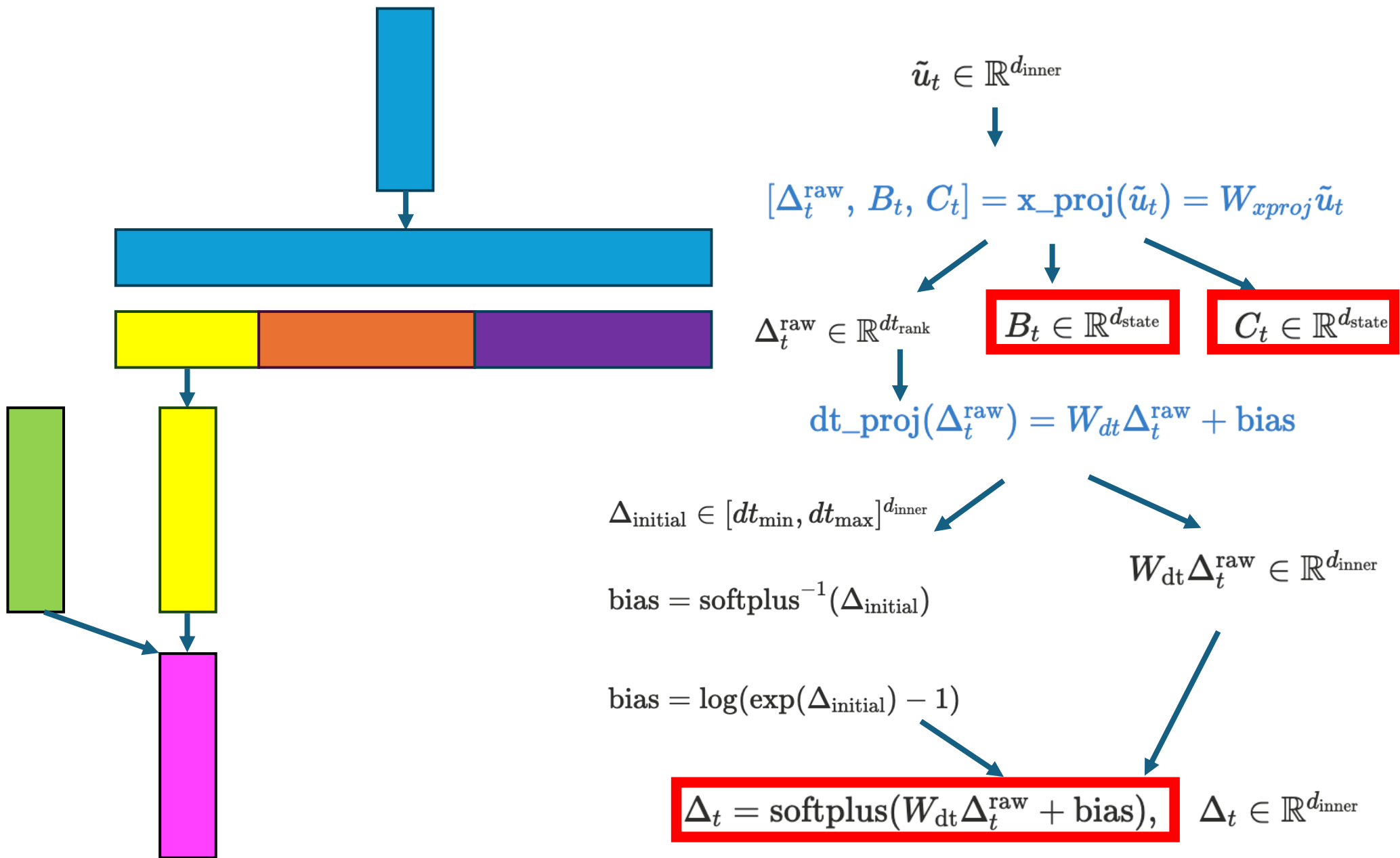




```
self.in_proj = nn.Linear(d_model, 2*d_inner)

self.conv1d = nn.Conv1d(
    in_channels=d_inner, out_channels=d_inner,
    kernel_size=d_conv, groups=d_inner, padding=d_conv-1
) # depthwise causal conv

self.act = nn.SiLU()
```

$$\begin{aligned}
 & x_t \in \mathbb{R}^{d\text{-model}} \\
 & \downarrow \text{in-proj}() \\
 & [u_t, z_t] \in \mathbb{R}^{d\text{-inner} \times 2} \\
 & \downarrow \text{split} \\
 & u_t \in \mathbb{R}^{d\text{-inner}}, z_t \in \mathbb{R}^{d\text{-inner}}
 \end{aligned}$$

$$\begin{aligned}
 & \text{SiLU}(\text{Conv}(\cdot)) \downarrow \\
 & \tilde{u}_t \in \mathbb{R}^{d\text{-inner}}
 \end{aligned}$$

$$\begin{aligned}
 & \text{x-proj}() \downarrow \\
 & [d\text{-row}, B_t, C_t] \in \mathbb{R}^{d\text{-rank} + 2 \times d\text{-sz} \times d\text{-sz}}
 \end{aligned}$$

$$\begin{aligned}
 & \text{split} \downarrow \\
 & d\text{-row} \in \mathbb{R}^{d\text{-rank}}, B_t \in \mathbb{R}^{d\text{-sz} \times d\text{-sz}}, \\
 & C_t \in \mathbb{R}^{d\text{-sz} \times d\text{-sz}}
 \end{aligned}$$

$$\begin{aligned}
 & d\text{-proj}() \downarrow \\
 & d\text{-inner} \in \mathbb{R}^{d\text{-inner}} \\
 & \downarrow \\
 & \Delta_t = \text{Softplus}(d\text{-inner} + \text{bias})
 \end{aligned}$$

$$\begin{aligned}
 \text{bias} &= \text{Softplus}'(\Delta) \\
 &= \log(\exp(\Delta) + 1)
 \end{aligned}$$